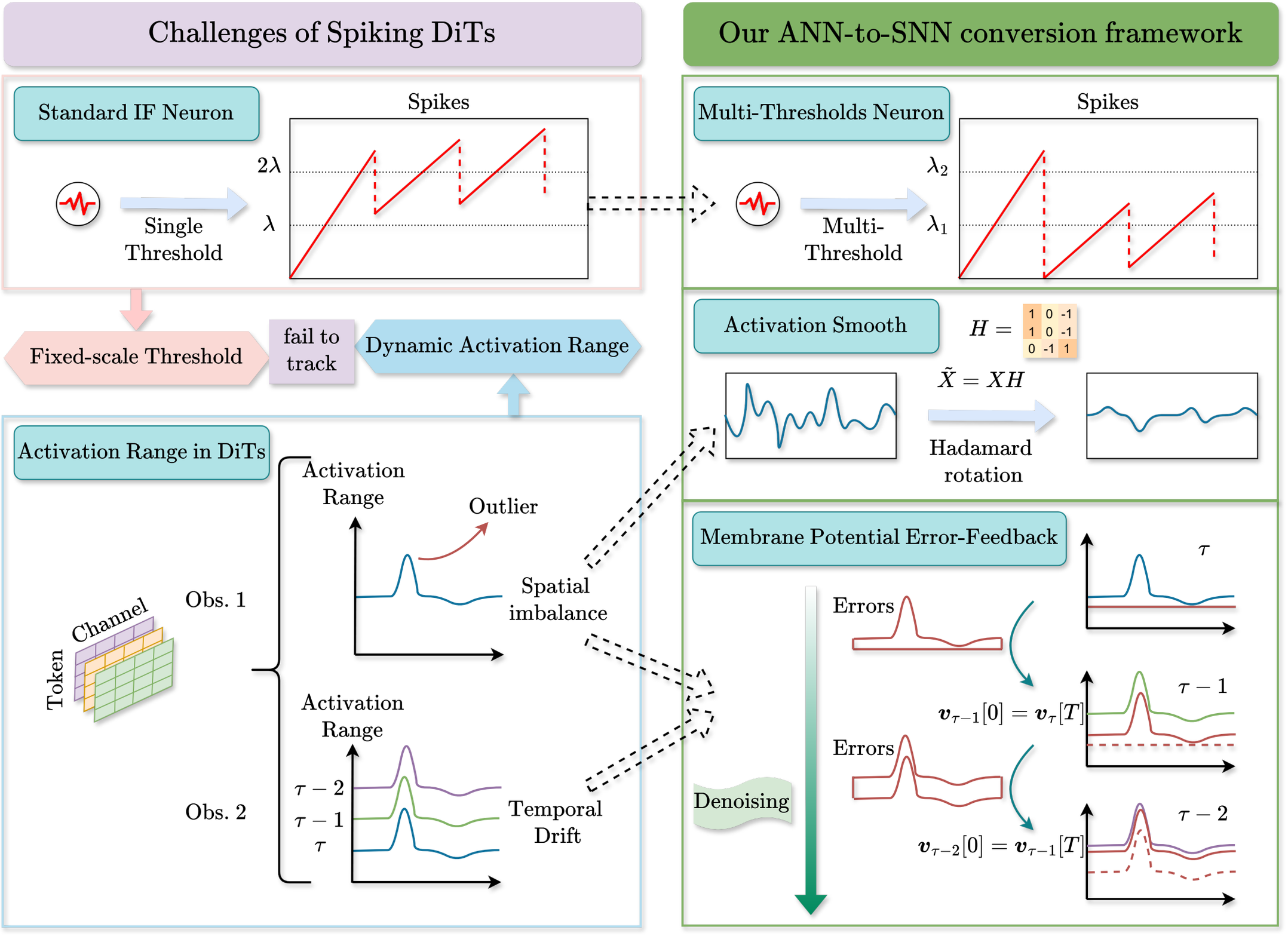

Challenges of Spiking DiTs
Our ANN-to-SNN conversion framework
Standard IF Neuron
Spikes
2λ
λ
Single Threshold
Multi-Thresholds Neuron
Spikes
λ_2
λ_1
Multi-Threshold
Fixed-scale Threshold
fail to track
Dynamic Activation Range
Activation Smooth
$H =$
1 0 -1
1 0 -1
0 -1 1
$\tilde{X} = XH$
Hadamard rotation
Activation Range in DiTs
Activation Range
Outlier
Spatial imbalance
Channel
Token
Obs. 1
Obs. 2
Activation Range
$\tau - 2$
$\tau - 1$
τ
Temporal Drift
Membrane Potential Error-Feedback
τ
Errors
$\tau - 1$
$\boldsymbol{v}_{\tau-1}[0] = \boldsymbol{v}_{\tau}[T]$
Errors
Denoising
$\tau - 2$
$\boldsymbol{v}_{\tau-2}[0] = \boldsymbol{v}_{\tau-1}[T]$